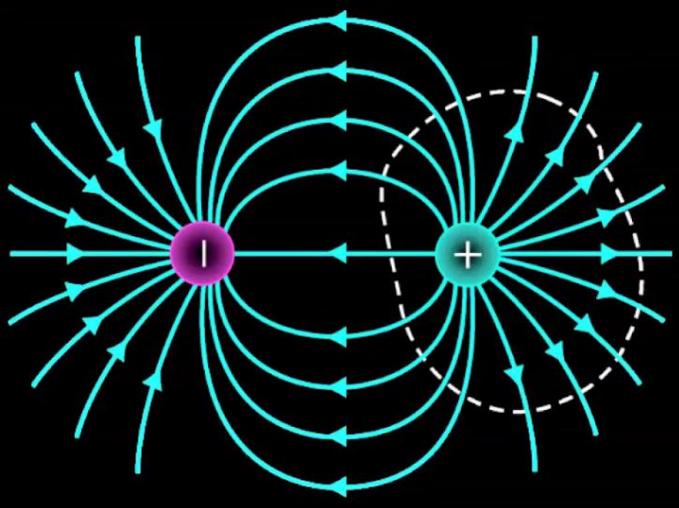
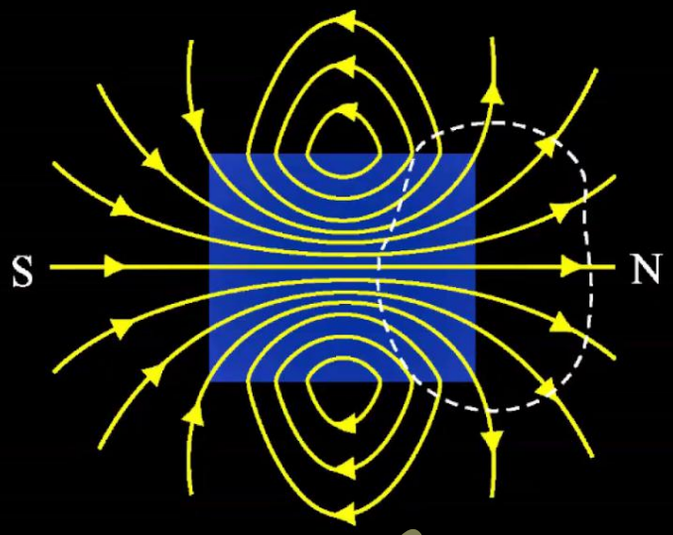
## Electric Dipole Vs Magnetic Dipole:




The above figure shows field lines for an electric dipole and a small solenoid. Although their field lines are not exactly same but for points outside the solenoid the field lines appear to be pretty similar to those of an electric dipole. (Though one essential difference is that in solenoid, magnetic field lines form closed loops and hence flux through closed Gaussian surface is zero unlike the case of electric dipole. Inside the solenoid, direction of magnetic field lines is opposite to that in electric dipole.

If we reduce the sizes of the dipole (or current loop) to a single point practically there is no "inside" and everything is outside. In such a scenario we can visualise that the field lines for two cases are almost same. Thus we might be able to draw some analogies between electric dipole and magnetic dipole (current loop).

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## Analogous Formulae For Magnetic Point Loop (Dipole) & Electric Dipole:

We find that not only the field lines of a point dipole and point current loop look alike, but many of the quantitative formulae describing their behaviours have same forms. For instance:

**Axial Field:** $\frac{\mu_0}{4\pi}\frac{2\vec{m}}{r^3}$ looks just like $\frac{1}{4\pi\epsilon_0}\frac{2\vec{p}}{r^3}$

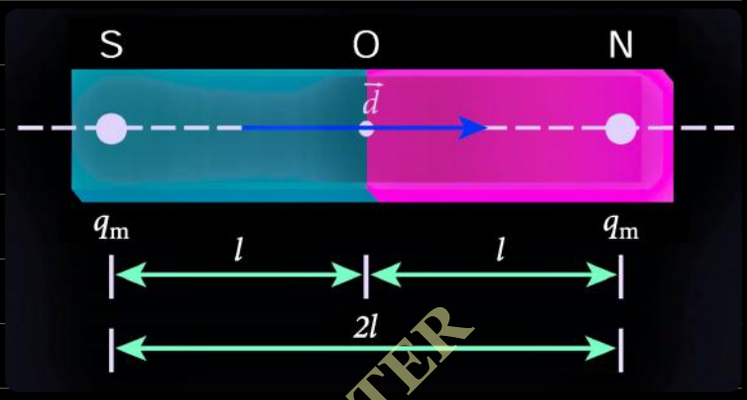
(Here $\vec{m} = i\vec{A}$ (current × Area Vector of the loop).

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**TORQUE:** $\vec{m} \times \vec{B}$ looks similar to $\vec{p} \times \vec{E}$

**POTENTIAL ENERGY:** $-\vec{m} \cdot \vec{B}$ looks similar to $-\vec{p} \cdot \vec{E}$

## COULOMB THEORY OF MAGNETISM:




The similarities listed in the previous section suggest that the behaviours of a current loop can be described by following Hypothetical (But inaccurate) model:

1.) There are two types of magnetic charges (poles) namely positive magnetic charge (north pole) $+q_B$ and negative magnetic charge (south pole) $-q_B$ such that a magnetic charge placed in a magnetic field $\vec{B}$ experiences a force $q_B\vec{B}$.

2. A magnetic charge $q_B$ produces a magnetic field given by

$$\vec{B} = \frac{\mu_0}{4\pi}\frac{q_B}{r^2}$$ at a distance $r$ from it and this field points radially outwards from $q_B$.

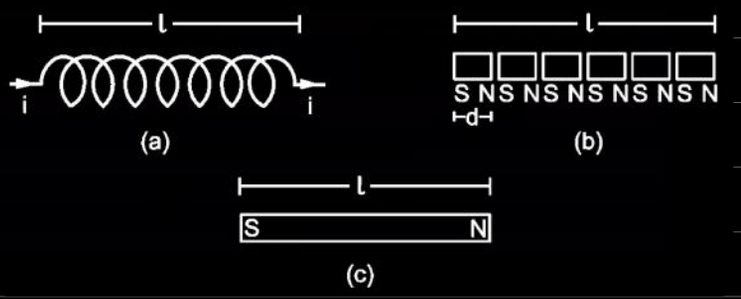
3. A magnetic dipole is formed when a negative magnetic charge $-q_B$ and a positive magnetic charge $+q_B$ are kept at a separation $\vec{l}$ and their magnetic dipole moment is $q_B\vec{l}$ with $\vec{l}$ pointing from $-q_B$ to $+q_B$. The line joining $-q_B$ to $+q_B$ is called axis of the dipole.

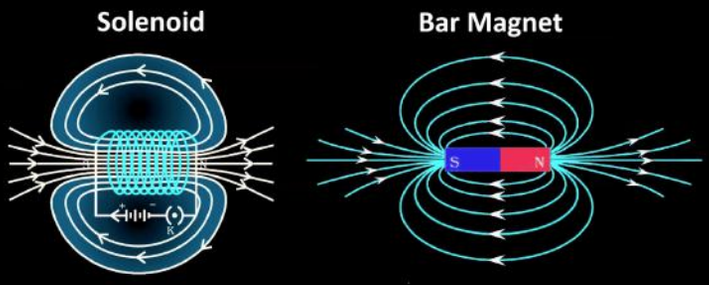
4. A current loop of area A and current i may be replaced by a magnetic dipole of dipole moment $\vec{m} = i\vec{A} = q_B\vec{l}$ with area vector pointing from $-q_B$ to $+q_B$.

→ Even though this model is inaccurate, it gives us reasonably correct results while working with permanent magnets due to its simplicity and therefore it is widely used.

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## Solenoid As A Bar Magnet:




A solenoid closely resembles combination of circular loops kept side by side. If $i$ is the current through the turns and $A$ be the area of the cross section then dipole moment of each turn is $iA$. Each such turn now can be replaced by a magnetic dipole with pole strength $q_B$ and separation $d$ such that $q_B d = iA$ and if $d$ is chosen in such a way that north pole of one dipole coincides with south pole of another, then north pole and south poles of adjacent dipoles cancel each other and we only have non-zero north poles and south poles at the ends of the solenoid. Thus a current carrying solenoid can be replaced by a single south pole and north pole of pole strength $q_B$ each placed at a separation equal to the length of the solenoid.

Once we have this model of the solenoid, we can use it for finding field due to the solenoid at any arbitrary point.

→ Accuracy of this model improves as the distance from solenoid becomes larger and larger as compared to dimensions of the solenoid.

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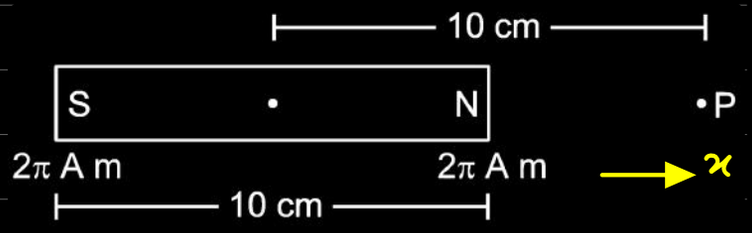
### Example:

**A solenoid of length 10 cm and radius 1 cm contains 200 turns and carries a current of 10 A. Find the magnetic field at a point on the axis at a distance of 10 cm from the centre.**

Solution: $m = NiA = 200 \times 10 \times \pi \times (0.01)^2$

$\Rightarrow m = 0.2\pi \text{ Am}^2 = q_B d$

$\Rightarrow q_B = \frac{m}{d} = \frac{0.2\pi}{0.1} = 2\pi \text{ Am}.$

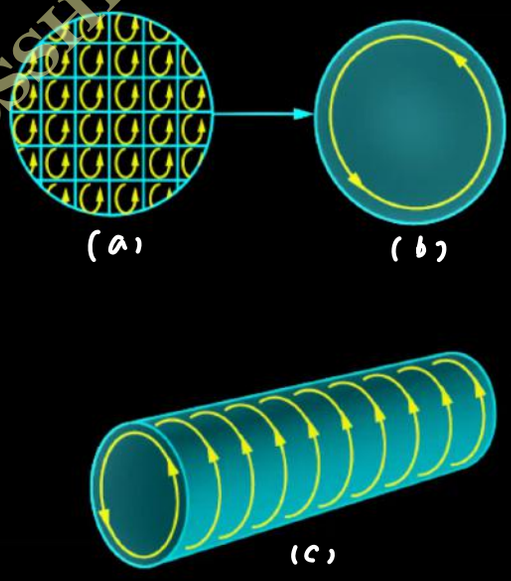



Now field due to north & South Poles:

$$B = \frac{\mu_0}{4\pi} \frac{2\pi}{(0.05)^2} \hat{i} - \frac{\mu_0}{4\pi} \cdot \frac{2\pi}{(0.15)^2} \hat{i} = 2.2 \times 10^{-4} \text{ T } \hat{i} \quad \text{Ans.}$$

(N: no of turns d: length of solenoid).

## MICROSCOPIC BEHAVIOUR IN A BAR MAGNET:




The figure (a) represents schematic section of an iron bar magnet. At microscopic level, atoms have microscopic currents due to circulation of electrons, these are represented by currents is small square loops. The edges of the loops that are shared between two squares have effectively zero current (equal and opposite currents cancel out) thus effective current is the current that runs along the periphery (i.e branches which are not shared between two loops). Thus each cross section of the bar magnet can be represented by a small surface current and accordingly the bar magnet as a whole acts just like a solenoid.
If $\lambda$ is the current per unit length of the bar magnet, A is its cross sectional area and $l$ is its length then its magnetic moment is given as: $\boxed{m = \lambda l A}$

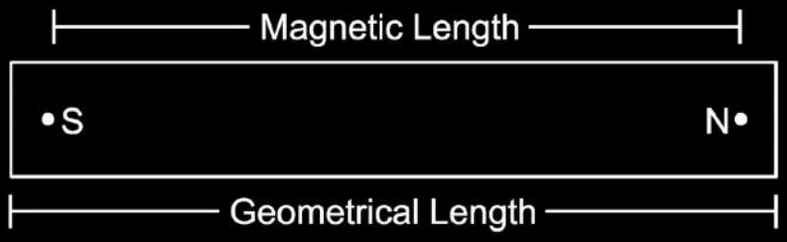
**MAGNETIC LENGTH Vs GEOMETRIC LENGTH:** If the bar magnet were modeled as magnetic charges at its two ends then maximum magnetic field would occur at the end points, however in practice the field declines near the ends and effective magnetic

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length is accordingly considered as 5/6 or 84% of the geometrical length.

Mathematically:

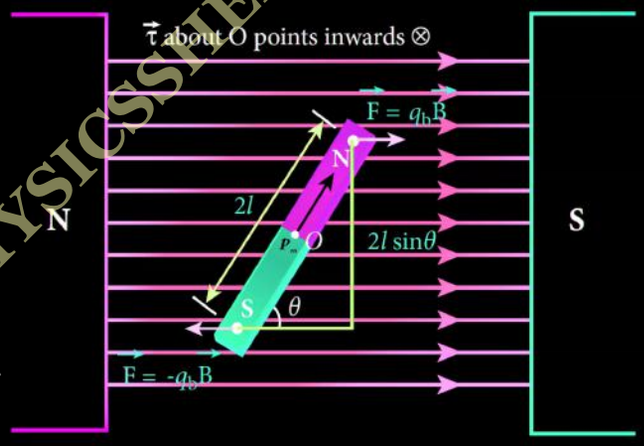
$$\frac{\text{Magnetic length}}{\text{Geometrical length}} = 0.84.$$




If we consider magnetic length as $2l$ and pole strength as $q_B$ then magnetic moment is given as $m = 2q_B l$.

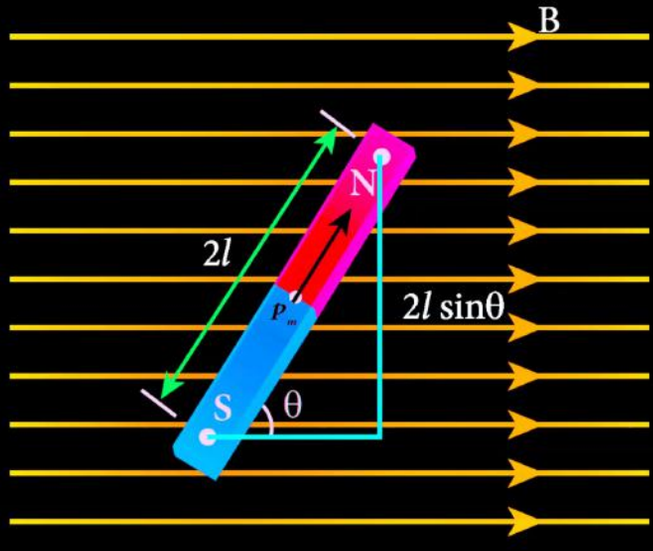
## Torque On A Bar Magnet In Uniform Magnetic Field:

Just like for the case of an electric dipole, the torque on a bar magnet is given as $\vec{\tau} = \vec{m} \times \vec{B}$. We are not writing a detailed derivation here, though student is encouraged to mentally verify this using the schematic in adjacent figure.




## Potential Energy Of A Bar Magnet:

Just like a dipole in electric field, the potential energy of a Bar Magnet in uniform magnetic field is given as $U = -\vec{m} \cdot \vec{B}$ or $U = -mB\cos\theta$. This can be easily derived by using the fact that magnetic potential energy is negative of the work done by magnetic torque thus, $dU = -\vec{\tau} \cdot d\vec{\theta}$ i.e $dU = -mB\sin\theta\, d\theta\, \cos(180°)$. We integrate this equation assuming that $U = 0$ for $\theta = 90°$ and thus get $U = -mB\cos\theta$.




→ Least PE occurs when magnet is parallel to external field

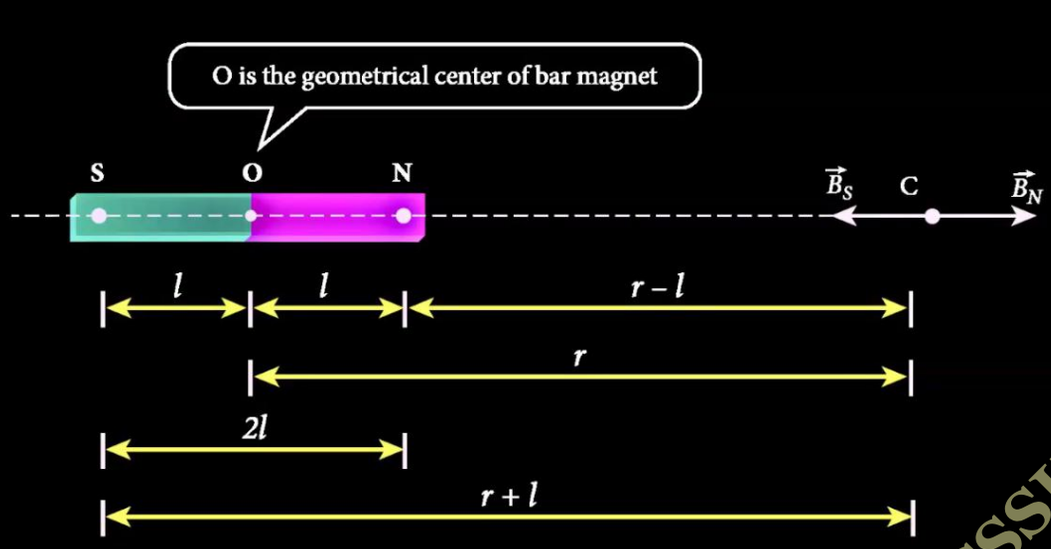
→ Maximum PE occurs when magnetic moment is antiparallel

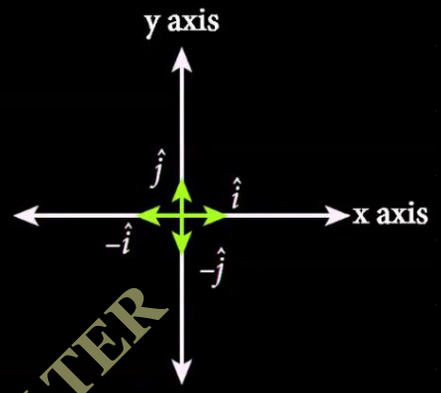
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to applied field.

## MAGNETIC FIELD DUE TO A BAR MAGNET:

### (A) END ON POSITION (AXIAL POSITION):




Consider a point C at a distance $r$ from mid point of a bar magnet as shown in the figure above. Using coulomb's law for magnetism:

$$\vec{B} = \vec{B}_N + \vec{B}_S = \frac{\mu_0}{4\pi}\frac{q_B}{(r-l)^2}\hat{i} + \frac{\mu_0}{4\pi}\frac{q_B}{(r+l)^2}(-\hat{i})$$

$$\Rightarrow \quad B = \frac{\mu_0}{4\pi}\cdot\frac{2r\,[2q_B\,l]}{(r^2-l^2)^2}\hat{i} = \frac{\mu_0}{4\pi}\frac{2r\vec{m}}{(r^2-l^2)^2} \quad \text{thus:}$$

$$\boxed{\vec{B}_{axis} = \frac{\mu_0}{4\pi}\frac{2r\vec{m}}{(r^2-l^2)^2}}$$

→ For $r \gg l$ this reduces to the familiar: $\vec{B}_{axis} = \frac{\mu_0}{4\pi}\frac{2\vec{m}}{r^3}$.

**EXAMPLE:** A magnet is 10cm long and its pole strength is 120 cgs units (1 cgs unit = 0.1 Am). Find the magnitude of magnetic field B at a point on its axis at a distance 20cm from it.

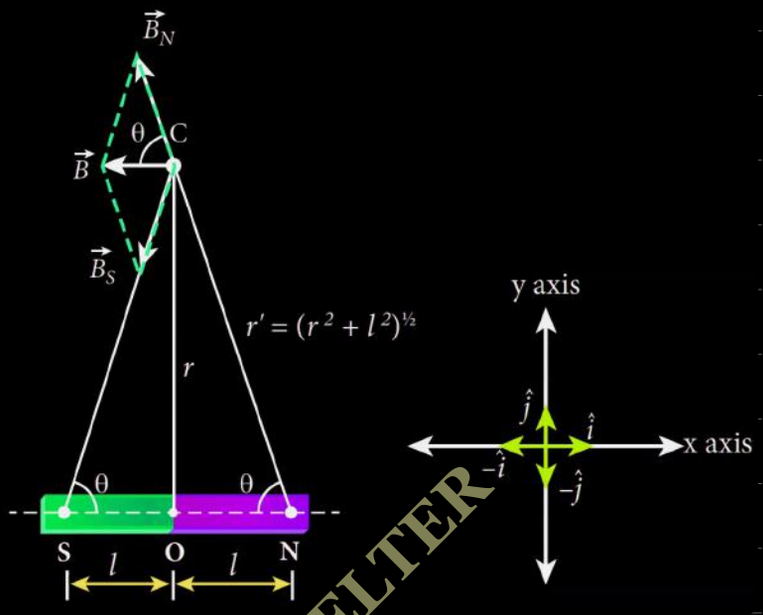
**Solution:**

$$\vec{B} = \frac{\mu_0}{4\pi}\cdot\frac{2r\vec{m}}{(r^2-l^2)^2} = \frac{10^{-7}\times 2\times 0.2\times 12\times 0.1}{(0.2^2-0.05^2)^2} = 3.41\times10^{-5}\,T$$

Thus $B = 3.4\times10^{-5}\,T = 0.34$ Gauss (G). [$10^4$ Gauss = 1 Tesla.]

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## (B) Broad Side On Position (Equatorial Position):




Consider a bar magnet of pole strength $q_B$ as shown in the figure. Here, magnetic field at C using Coulomb's law and symmetry is given as:

$$-\frac{\mu_0}{4\pi}\frac{q_B}{(r^2+l^2)}\cdot 2\cos\theta\,\hat{i}$$

Also $\cos\theta = l/\sqrt{r^2+l^2}$ thus:

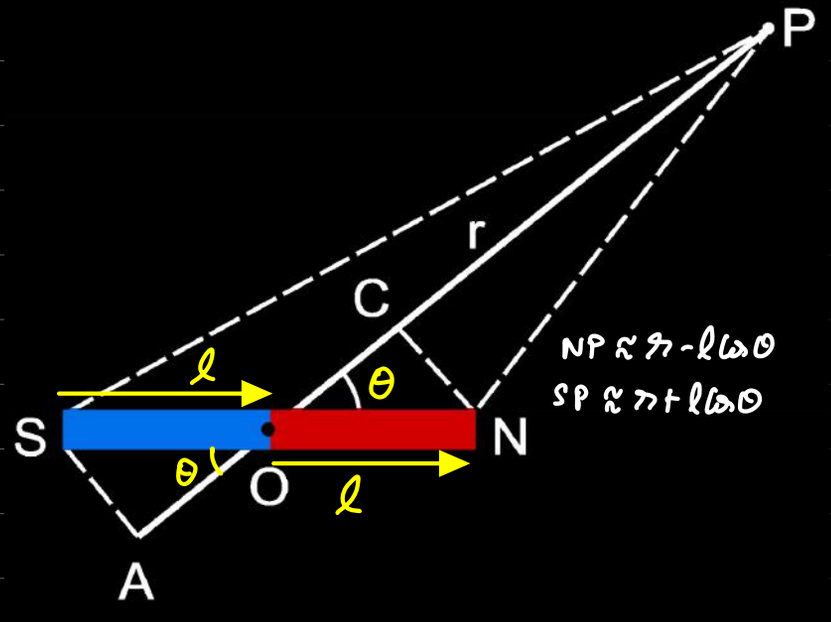
$$\vec{B} = -\frac{\mu_0}{4\pi}\frac{2q_B l}{(r^2+l^2)^{3/2}}\,\hat{i}$$

$$\Rightarrow \boxed{\vec{B} = -\frac{\mu_0}{4\pi}\frac{\vec{m}}{(r^2+l^2)^{3/2}}}$$

→ For $r >> l$, this reduces to the familiar formula $\vec{B} = \frac{-\mu_0 \vec{m}}{4\pi r^3}$.

## Magnetic Scalar Potential:

We define magnetic scalar potential just as we defined the potential for electric fields. Considering the adjacent figure, potential at P is given as (Assuming $r >> l$):




$$\frac{\mu_0}{4\pi}\left(\frac{q_B}{r - l\cos\theta}\right) - \frac{\mu_0}{4\pi}\frac{q_B}{(r + l\cos\theta)}$$

Thus:

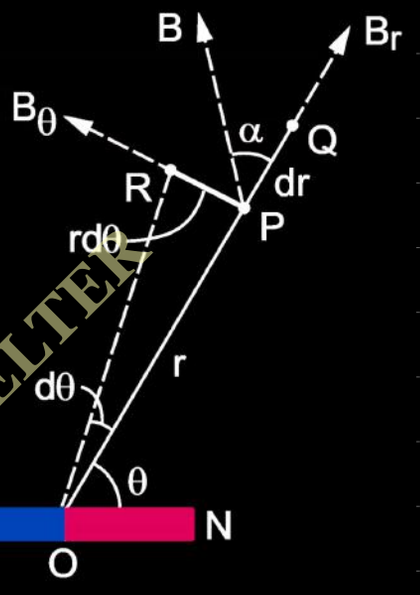
$$V = \frac{\mu_0}{4\pi}\left(\frac{2q_B l\cos\theta}{r^2 - l^2\cos^2\theta}\right) = \frac{\mu_0}{4\pi}\frac{m\cos\theta}{r^2 - l^2\cos^2\theta}$$

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Now we can further approximate this if $r \gg l$ as:

$$V \approx \frac{\mu_0}{4\pi} \frac{m \cos\theta}{r^2}$$

## Radial And Tangential Fields Due To Point Dipole:




Just like the case of electric dipoles, we can write:

$$B_r = -\frac{\partial V}{\partial r} \quad \text{and} \quad B_\theta = -\frac{1}{r}\frac{\partial V}{\partial \theta}$$

thus:

$$B_r = \frac{\mu_0}{4\pi} \cdot \frac{2m}{r^3} \cos\theta \quad \text{and}$$

$$B_\theta = \frac{\mu_0}{4\pi} \frac{m}{r^3} \sin\theta \quad \text{and the}$$

angle of resultant field with radial line is given as:

$\tan\alpha = B\theta / B_r = \tan\theta/2$ thus:

$$\tan\alpha = \frac{\tan\theta}{2}$$

→ In many problems we are required to find $\theta$ for which field is perpendicular to the dipole. Here, $\alpha + \theta = 90°$ thus: $\tan(90° - \theta) = \frac{\tan\theta}{2}$

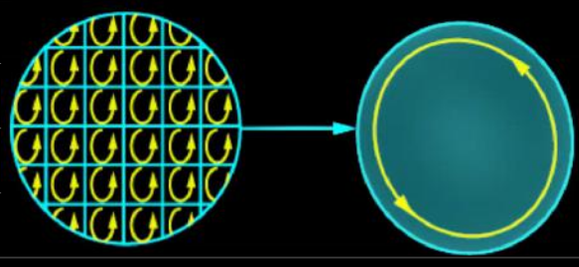
$\Rightarrow$ $\tan\theta = \sqrt{2}$

→ Net magnitude of magnetic field is $\sqrt{B_\theta^2 + B_r^2}$ thus:

$$B = \frac{\mu_0}{4\pi} \frac{m\sqrt{1+3\cos^2\theta}}{r^3}$$

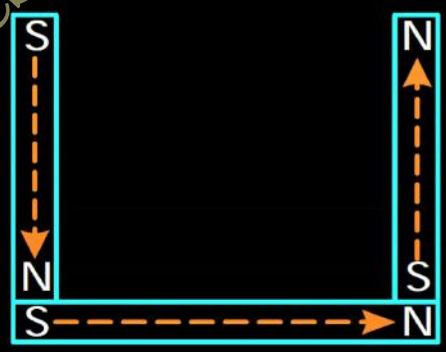
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# Magnetic Moments Of Pieces Of A Bar Magnet:

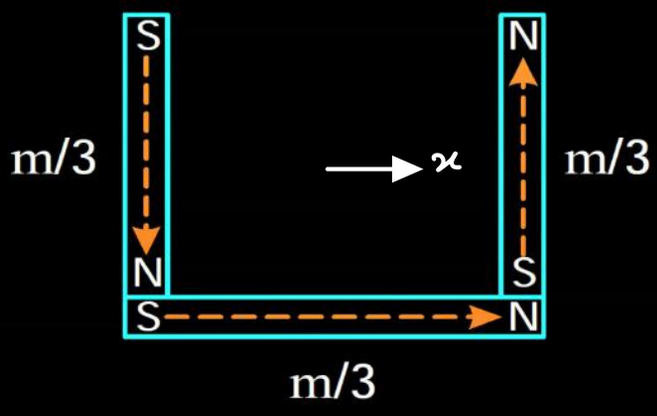


Recall that if $\lambda$ is surface current per unit length then magnetic moment of a piece of cross section A and length $l$ is simply $\lambda \times Al = \lambda \times \forall$ where V is volume of the piece. Thus magnetic moment of a piece is directly proportional to its volume. $\lambda$ just depends on density of molecular currents and for a uniformly magnetized material, $\lambda$ is constant. Thus if $\vec{m}$ is net magnetic moment of a bar magnet, an $\eta$ fraction of its volume will have a magnetic moment $\eta\vec{m}$.

**Example:** A bar magnet of magnetic moment $\vec{m}$ is bent in to a U shape comprising of three identical segments as shown in adjacent figure. Find the net magnetic moment of the resultant geometry.

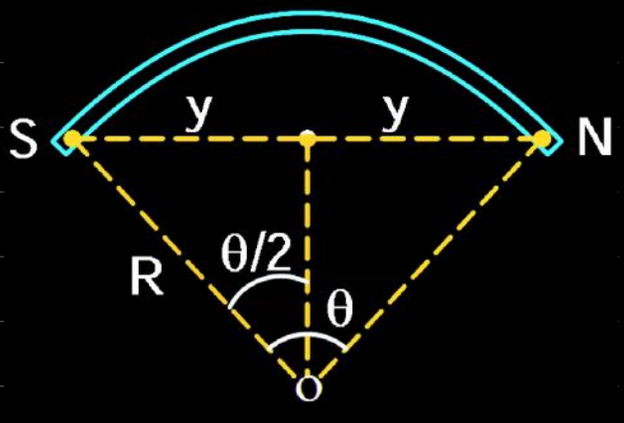



**Solution:** Based on our discussion in previous section, we can see that each part has a magnetic moment $m/3$. Thus taking vector sum of magnetic moments of each part, the net magnetic moment becomes $\frac{m}{3}$ î Ans.




**Example:** A bar magnet of length $l$ and magnetic moment $m$ is bent in to an arc of angle $\theta$. What is the magnetic moment of this new geometry?

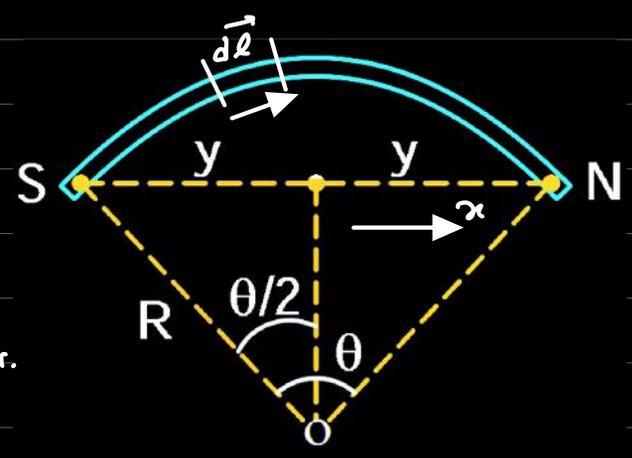
**Solution:** Consider a length $\vec{dl}$ element of the arc as shown in the figure.



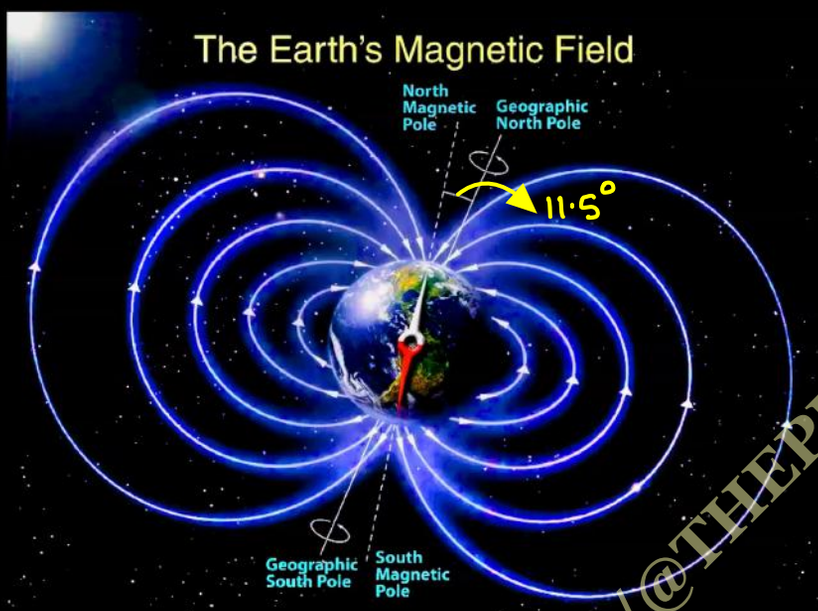

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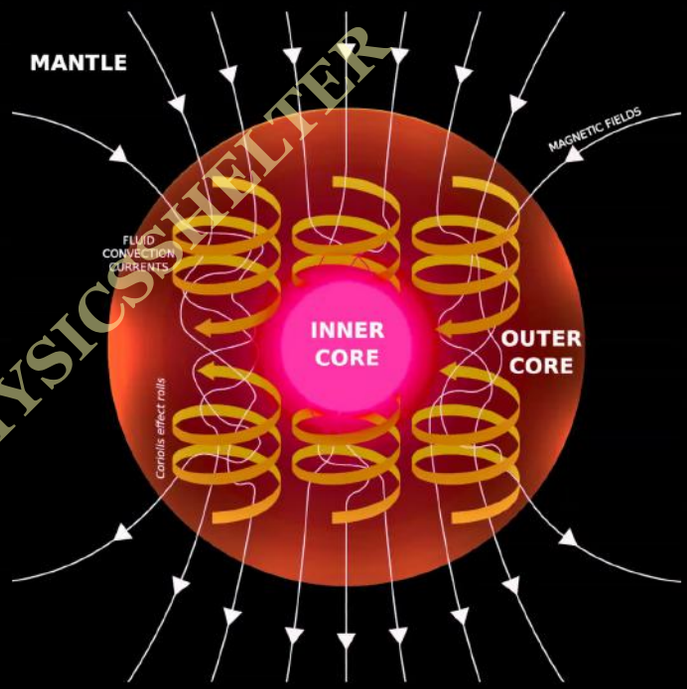
The magnetic moment of this is : $d\vec{m} = \frac{m}{\ell}\, d\vec{\ell}$. Thus net magnetic moment is simply $\vec{m} = \frac{m}{\ell} \cdot 2R\sin\frac{\theta}{2}\,\hat{i}$

also $\ell = R\theta$ thus, $\vec{m} = \frac{m \sin\theta/2}{\theta/2}\,\hat{i}$ Ans.




## EARTH'S MAGNETIC FIELD:




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Earth's magnetic field is believed to be generated by electric currents in the conductive iron alloys of its core created by convection currents due to heat escaping from the core. The heat is generated due to loss in potential energy of heavier elements sinking to the core as well as radioactive decay of the elements in the core. The flow pattern occurs due to rotation of earth (Coriolis effect). For simplistic calculations, we may approximate earth's magnetic field to be generated by a magnetic dipole of magnetic moment $8 \times 10^{22}$ J/T located at the centre of the earth (see figure). The axis of the dipole makes an angle of $11.5^\circ$ with earth's axis of rotation. The axis of the dipole cuts the surface of earth at two points, one near the geographical north pole and the other near the geographical south pole. These points are known as geomagnetic North

pole and Geomagnetic south poles respectively. (Carefully notice the polarity of the magnetic dipole embedded in earth).

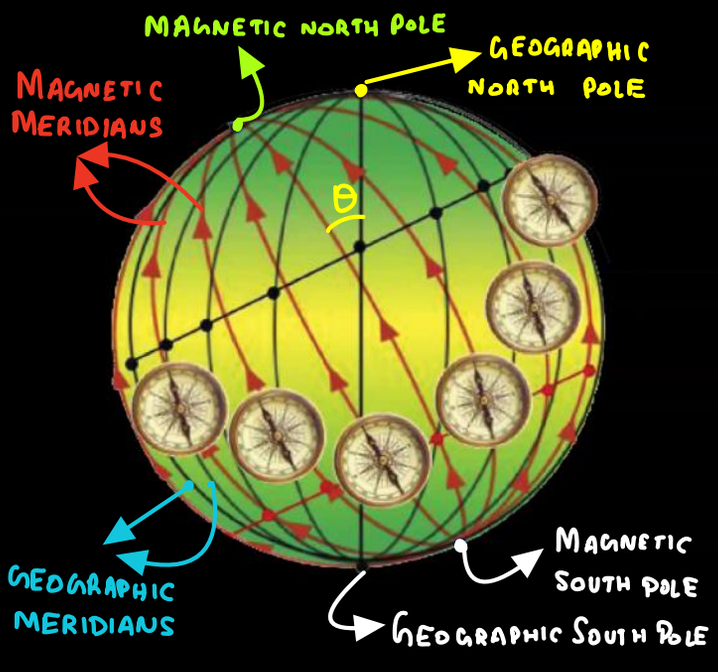
If we assume the embedded magnetic dipole embedded in earth as a pair of north and south poles, its south pole will be towards Geomagnetic north and its North pole will be towards Geomagnetic South.

Earth's magnetic field changes both in direction and magnitude as the time passes. Studies of magnetic rocks have revealed that magnetic field of earth has reversed its direction 171 times in past $7.6 \times 10^7$ years. (The latest reversal is believed to have occured 10,000 years ago).

**ELEMENTS OF EARTH'S MAGNETIC FIELD:** For the purpose of Geomagnetic studies, the magnetic field of earth at any point on surface is characterised by three parameters namely angle of dip, angle of declination and horizontal component of earth's magnetic field. These are described in the following sub-sections.

**MAGNETIC MERIDIAN:**

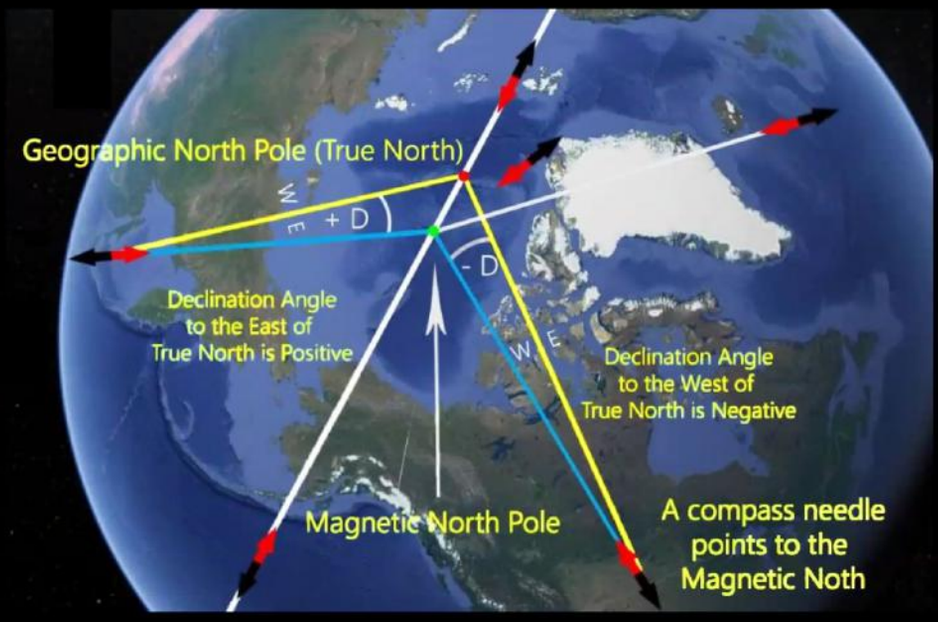
At any point on surface of earth, the plane containing magnetic North pole, Magnetic south pole and that point is called magnetic meridian. The intersection of magnetic meridian with earth's surface is a kind of "Magnetic longitude". When a compass needle free to move in horizontal plane is placed on surface, it points along magnetic meridian.

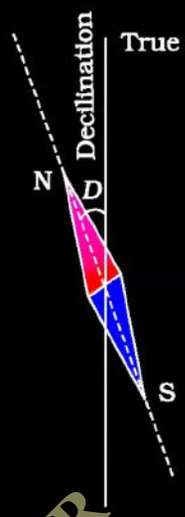



**GEOGRAPHIC MERIDIAN:** A plane containing geographic north pole, geographic south pole and the point on the surface defines the geographic meridian plane at that point. Its intersection with surface of earth is the longitude.

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## Angle Of Declination:

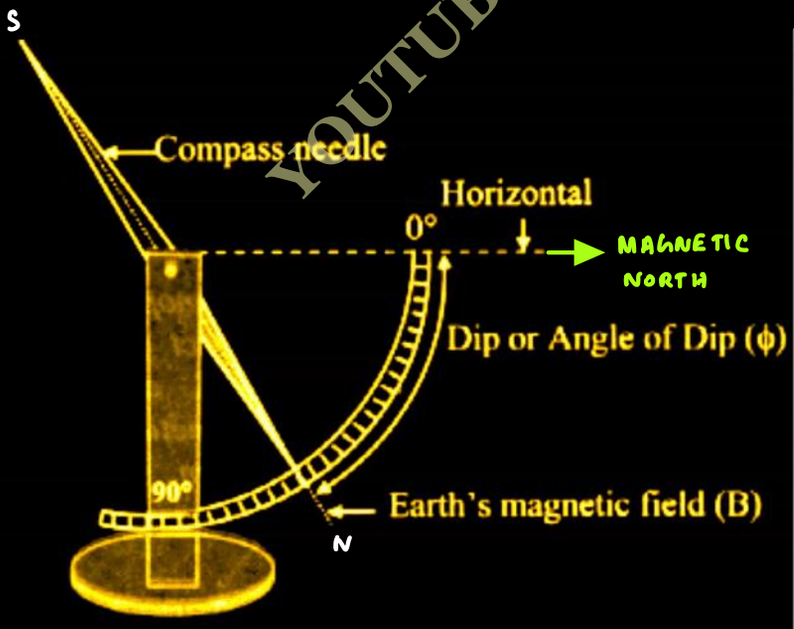


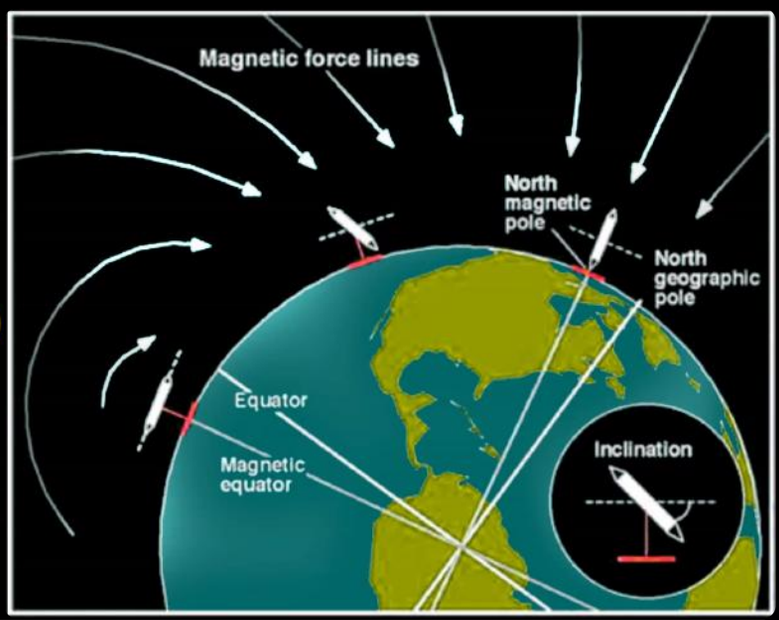




A magnetic needle free to move in horizontal plane, points toward the magnetic north-south direction.

Angle between the magnetic meridian and the geographic meridian at any point on surface of earth is called angle of declination. Angle of declination is considered as positive if magnetic north is to the east of geographic north and negative if magnetic north is to the west of geographic north. (In simple words, declination is angle between true north direction and magnetic north direction as indicated by a compass needle).

## Angle Of Dip: (or Inclination):

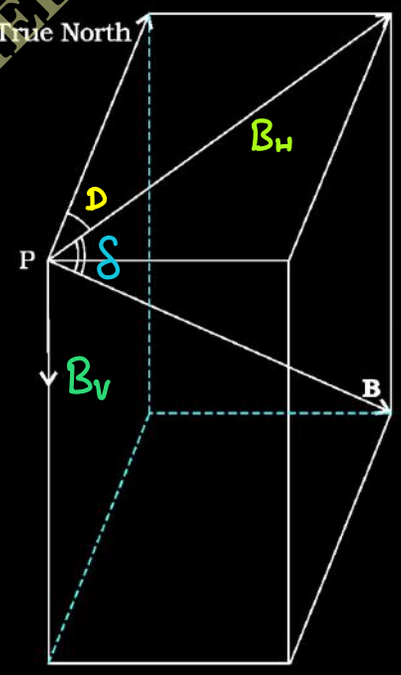



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The angle by which magnetic field at a location on earth points below the horizontal is known as angle of dip. This is also the angle by which a compass needle "dips" when placed such that it is free to rotate in the magnetic meridian (vertical plane). The angle of dip is zero at magnetic equator and 90° at the magnetic North pole. In magnetic southern hemisphere, instead of dipping, the magnetic needle points above the horizontal, thus there instead of dip we call it angle of inclination (or negative dip).

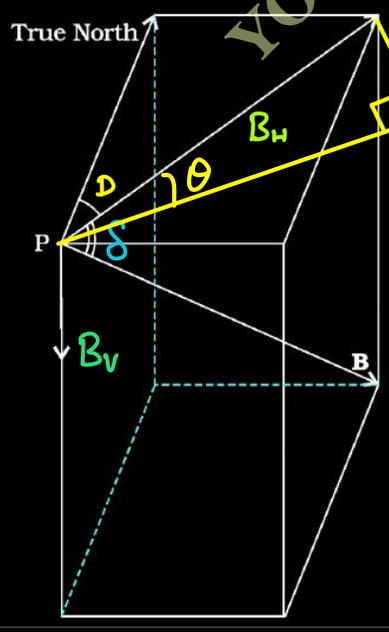
## Horizontal Component Of Earth's Magnetic Field:



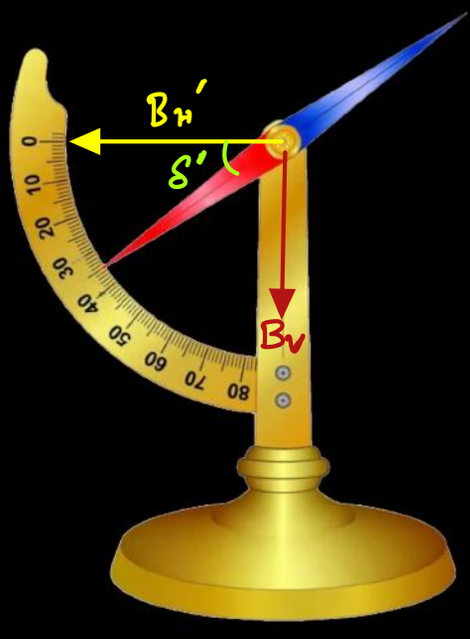

As the name suggests, it is the horizontal component of earth's magnetic field ($B_H$ in the adjacent figure). As evident from the figure, angle of Dip is the angle between net field vector and $B_H$. Thus $\boxed{B_H = B\cos\delta}$ where

$\delta$ is the angle of dip. Vertical component is accordingly $B_V = B\sin\delta$. Angle of declination (D in the figure) is the angle between true north and the $B_H$.

## Apparent Dip:




$B_H' = B_H\cos\theta$




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When a compass needle is free to rotate in a vertical plane other than the magnetic meridian then the angle through which the magnetic needle dips is called angle of apparent dip ($\delta'$).
If $B_H'$ is the component of horizontal magnetic field in this plane and $B_V$ is the vertical component of field then,

$$\tan\delta' = B_H'/B_V = B_H\cos\theta/B_V \quad (1) \quad (\because \theta \text{ is angle between magnetic meridian and the vertical plane of compass}).$$

Also for true dip:

$$\tan\delta = B_H/B_V \quad (2)$$

From (1) & (2),

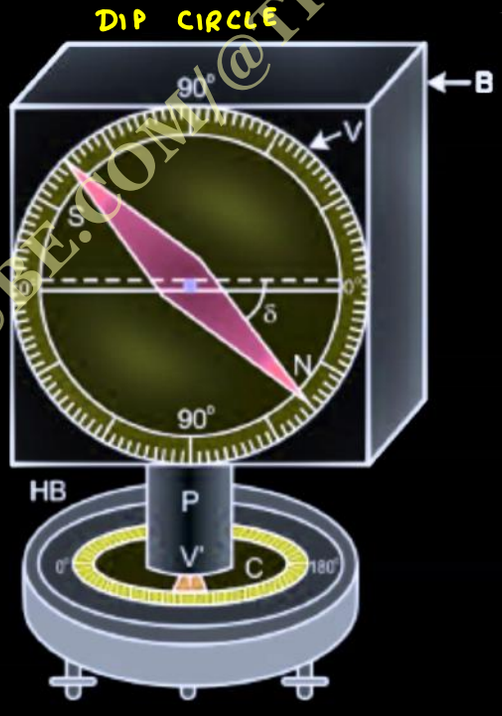
$$\frac{\tan\delta'}{\tan\delta} = \cos\theta \Rightarrow \boxed{\tan\delta' = \tan\delta\cos\theta}$$

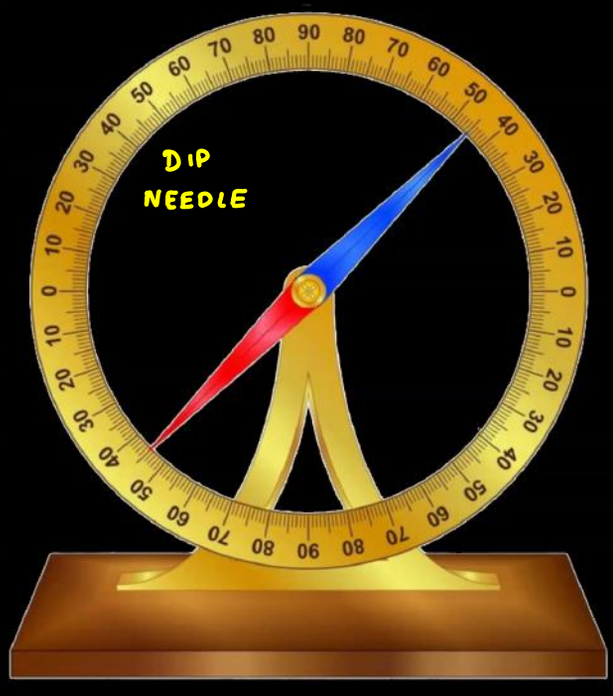
$\delta'$: Apparent dip  $\delta$: True dip  $\theta$: Angle between magnetic meridian and compass plane (vertical).

→ Apparent dip is 90° in plane perpendicular to magnetic meridian.

## Dip Circle:






A dip circle is an instrument which is used for measuring dip or inclination at any point on earth. It comprises of a compass needle mounted to rotate freely in a vertical plane and the direction of the vertical plane can be adjusted by rotating it about a

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pivot (see figure). To find the dip angle we must first locate the magnetic meridian. This is done by rotating the vertical plane until the angle of apparent dip becomes 90°. This gives us the plane perpendicular to the magnetic meridian. From here the vertical plane is rotated exactly 90° to place it in magnetic meridian and angle of dip is then read directly from the compass dial.

(In absence of dip circle, we may do the same procedure with a dip needle. This would require making manual markings for plane where apparent dip becomes zero). Dip circles are also provided with spirit levels and adjustment screws to ensure proper horizontal leveling of the instrument.

**EXAMPLE:** In a certain vertical plane, the dip circle records an apparent dip $\delta_1$, when the vertical dial is rotated through 90°, the recorded apparent dip is $\delta_2$. What is the true dip angle.

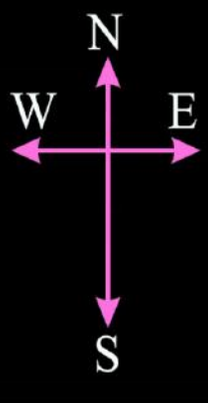
**Solution:** $\cot\delta' = \cot\delta \cos\theta$, thus $\cot\delta_1 = \cot\delta\cos\theta$ and $\cot\delta_2 = \cot\delta\cos(90°-\theta)$ i.e $\cot\delta\sin\theta$ thus:

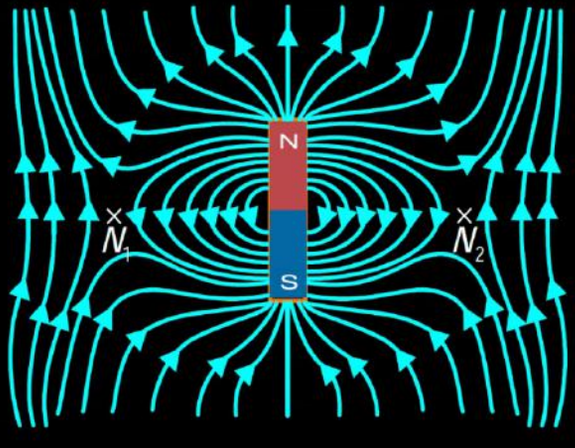
$\cot^2\delta_1 + \cot^2\delta_2 = \cot^2\delta \Rightarrow \delta = \cot^{-1}[\cot^2\delta_1 + \cot^2\delta_2]^{1/2}$ Ans.

## FIELD LINES AROUND A BAR MAGNET IN VARIOUS CONFIGURATIONS:

### CASE I: NORTH OF MAGNET TOWARDS GEOMAGNETIC NORTH POLE:

The magnetic field lines for the case when north pole of bar magnet points towards magnetic North pole are shown in adjacent figure. Here $N_1$ and $N_2$ are neutral points (Null points on Zero magnetic field points).



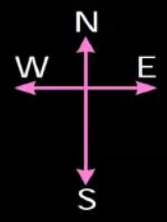




Here $N_1$ and $N_2$ are in the equatorial plane of the bar magnet (In the equatorial plane, field of magnet is against its dipole moment).

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## Case 2: North Pole Of Magnet Pointing Towards Geomagnetic South Pole:

In this case, south pole of the magnet is pointing towards Geomagnetic North pole and null points are obtained on the axis of the bar magnet (see $N_1$ and $N_2$ in the figure). Here the axial field of the bar magnet is against the horizontal component of earth's magnetic field.

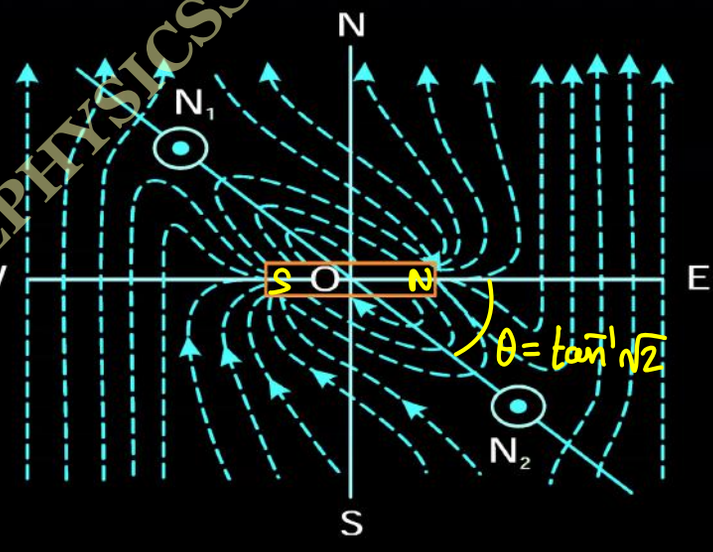




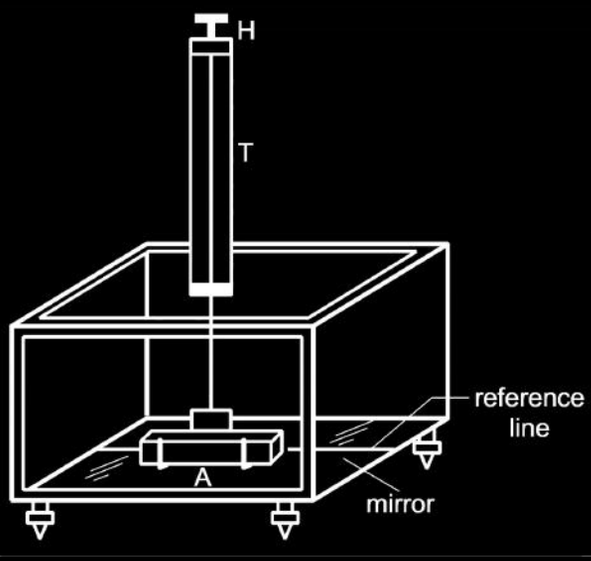


## Case III: Bar Magnet ⊥ To The Magnetic Meridian:

Recall that the magnetic field due to a bar magnet is perpendicular to it at an angle of $\tan^{-1}\sqrt{2}$ from its axis. Thus if bar magnet is aligned perpendicular to magnetic meridian as shown in the figure, the neutral points $N_1$ and $N_2$ are located on the line inclined at $\tan^{-1}\sqrt{2}$ to the axis of bar magnet.



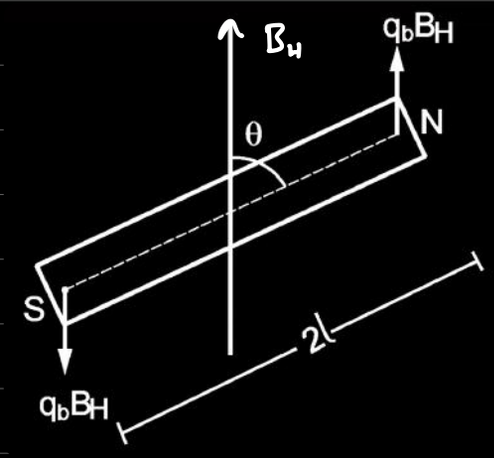

## Oscillation Magnetometer:






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Oscillation magnetometer is a device which measures the product of pole strength and horizontal component of earth's magnetic field. Here a magnet is suspended from a fine thread (negligible torsional constant) and made to undergo rotational oscillations in a horizontal plane under influence of magnetic torque of earth's field. When magnet is inclined at an angle $\theta$ with horizontal component of earth's field, it experiences a restoring torque $2q_b l B_H \sin\theta$ (see figure). Thus the torque is $mB_H \sin\theta$. Now if $I$ is the moment of inertia of the magnet about rotation axis then,




$$mB_H \sin\theta = -I\, d^2\theta/dt^2 \approx mB_H\theta \quad \text{(if } \theta \text{ is small)}.$$

Thus time period of oscillation is given as:

$$T = 2\pi\sqrt{\frac{I}{mB_H}} \qquad (1)$$

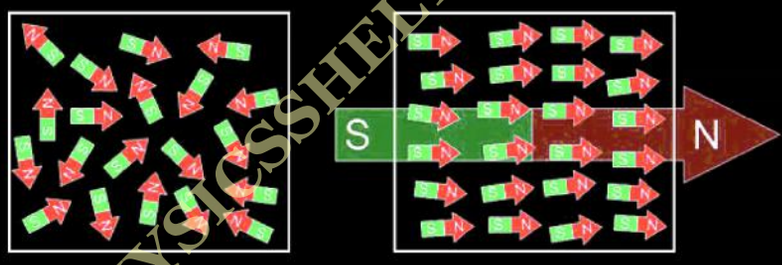
Time period of magnet's oscillation is measured using a stop watch. Mass is measured using a balance and dimensions are measured using mm scale. If the mass is M and dimensions perpendicular to rotation axis are a x b then moment of inertia is given as $I = M(a^2+b^2)/12$. Thus using equation (1) we can find the product $mB_H$ and if length of magnet is known, its pole strength can be found using the formula $m = 2q_B l$. We could also find the value of $B_H$ if we have a bar magnet of known $m$.

**MAGNETIZATION OF MATERIALS:** As you might have studied in your chemistry syllabus, electrons in matter have spin and orbital magnetic moments, thus these electrons also produce magnetic fields. Since the orientation of various atoms in matter is generally arbitrary, at macroscopic level we don't see any magnetic field in general. (Besides nuclei also have their own magnetic moments).

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However when materials are kept in external magnetic field, the microscopic magnetic dipoles at atomic levels experience magnetic torques, leading to some kind of alignment tendency of the dipoles. The alignment is only partial because of thermal agitation tends to randomize the magnetic moments. The degree of alignment increases with reduction in temperature and increasing the strength of applied field. With sufficiently large fields, the alignment is nearly perfect and we say that material is magnetically saturated.

## MAGNETIZATION VECTOR: ($\vec{M}$):




As just described, the materials develop a net non zero magnetic moment when subjected to an external magnetic field. The net magnetic moment induced per unit volume is called intensity of magnetization ($M$). Thus, if $\vec{m}$ is the net magnetic moment of a sample and $\forall$ is it volume then its magnetization vector is given as:

$$\vec{M} = \frac{\vec{m}}{\forall}$$

→ Magnetization is also sometimes known as intensity of magnetization.

**EXAMPLE:** A bar magnet has magnetic moment $2.5\,Am^2$ and a mass $6.6 \times 10^{-3}$ Kg. If density of steel is $\rho = 7.9 \times 10^3\ Kg/m^3$ find the intensity of magnetization.

**Solution:** $M = m/\forall = m/(m_0/\rho)$ where $m_0$ is the mass.

Thus $M = \rho m/m_0 = \dfrac{7.9 \times 10^3 \times 2.5}{6.6 \times 10^{-3}}\ A/m = 3 \times 10^6\ A/m$ Ans.

**$\vec{H}$ FIELD (MAGNETIC INTENSITY, MAGNETISING FIELD):** While dealing with magnetic materials, if $\vec{B}_0$ is the applied magnetic field for magnetising a sample, it is often convenient to define a quantity $\vec{B}_0/\mu_0$ which has same dimensions as Magnetisation and is alternatively

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knows as Magnetic Intensity or Magnetizing field ($\vec{H}$). Thus

$$\vec{H} = \vec{B_0}/\mu_0$$

**Susceptibility: ($\chi$):** The net magnetization of a material is proportional to the magnetizing field for most materials. This constant of proportionality is called susceptibility ($\chi$). Thus:

$$\vec{M} = \chi \vec{H}$$

**Field Due To Magnetization:** Imagine sample of some material kept in a solenoid carrying a current $i$ and $n$ number of turns per unit length. Then the applied field is simply $B_0 = \mu_0 ni$ (i.e $H = ni$). In response to this a magnetization $\vec{M}$ occurs in the sample then it can be shown that field arising due to magnetisation of the sample is $\vec{B_m} = \mu_0 \vec{M}$. Now net field in the sample will be given as $\vec{B} = \vec{B_0} + \vec{B_m} = \mu_0 \vec{H} + \mu_0 \vec{M} = \mu_0 (1+\chi) \vec{H}$

Thus:

$$\vec{B} = \mu_0 (1+\chi) \vec{H} = (1+\chi) \vec{B_0}$$

**Relative Permeability And Permeability:** The term $1+\chi$ is called relative permeability and $\mu_0(1+\chi)$ is called permeability thus:

$$\mu_r = 1+\chi \quad \text{and} \quad \mu = \mu_0 \mu_r$$

Thus we can write the net field in the material as $\vec{B} = \mu \vec{H}$. This tells us that more the permeability more is the net magnetic field in the material for the same magnetizing field.

**Classification Of Materials Based On Susceptibility:**

Based on susceptibility the magnetic materials may be classified as Paramagnetic ($\chi > 0$) or diamagnetic ($\chi < 0$) or ferromagnetic ($\chi >> 1$). We will look in to physical phenomena behind these behaviors in later sections. Here we present a small table of some

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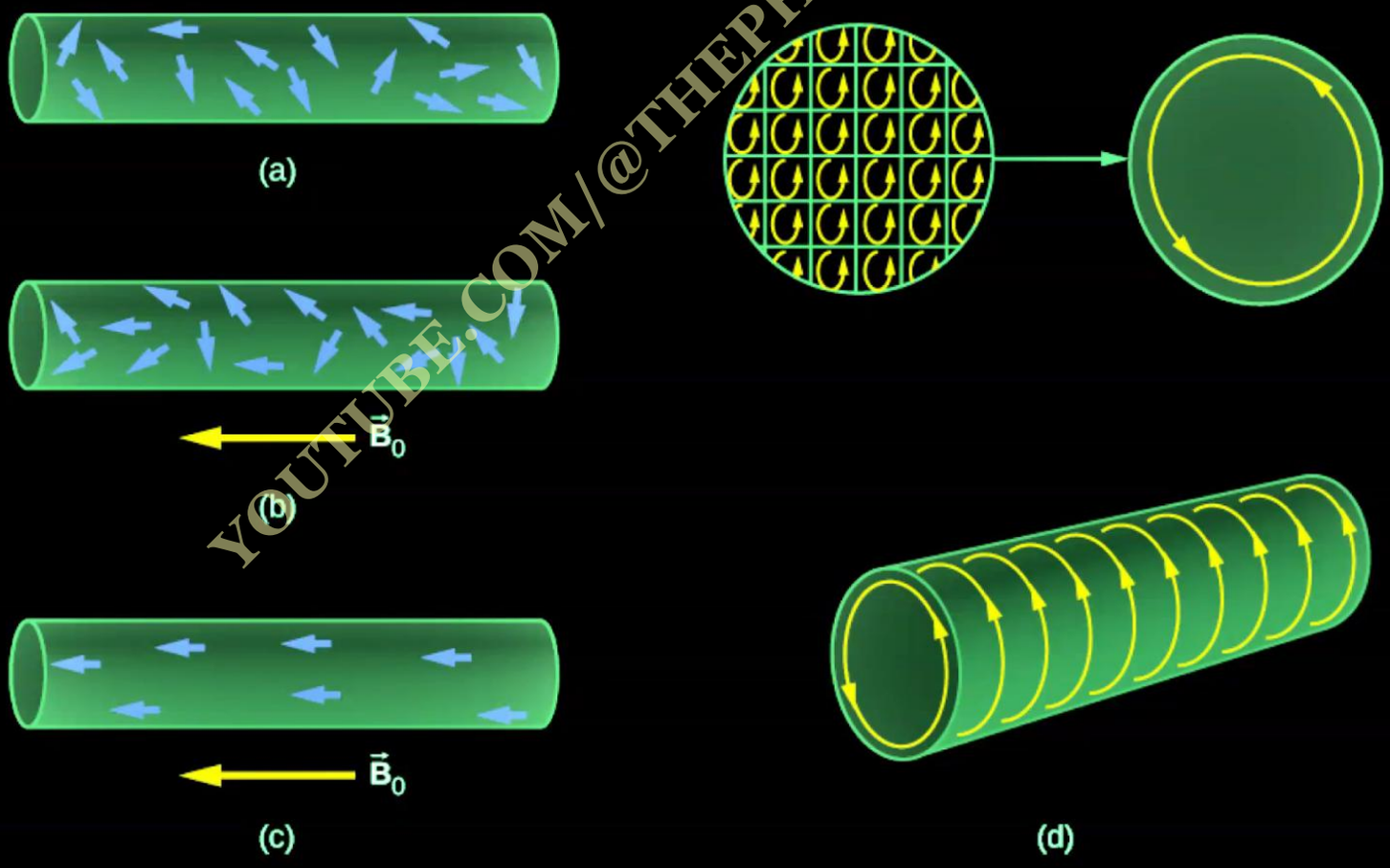
## paramagnetic and diamagnetic materials:

**MAGNETIC SUSCEPTIBILITY OF SOME ELEMENTS AT 300 K**

| Diamagnetic substance | $\chi$ | Paramagnetic substance | $\chi$ |
|---|---|---|---|
| Bismuth | $-1.66 \times 10^{-5}$ | Aluminium | $2.3 \times 10^{-5}$ |
| Copper | $-9.8 \times 10^{-6}$ | Calcium | $1.9 \times 10^{-5}$ |
| Diamond | $-2.2 \times 10^{-5}$ | Chromium | $2.7 \times 10^{-4}$ |
| Gold | $-3.6 \times 10^{-5}$ | Lithium | $2.1 \times 10^{-5}$ |
| Lead | $-1.7 \times 10^{-5}$ | Magnesium | $1.2 \times 10^{-5}$ |
| Mercury | $-2.9 \times 10^{-5}$ | Niobium | $2.6 \times 10^{-5}$ |
| Nitrogen (STP) | $-5.0 \times 10^{-9}$ | Oxygen (STP) | $2.1 \times 10^{-6}$ |
| Silver | $-2.6 \times 10^{-5}$ | Platinum | $2.9 \times 10^{-4}$ |
| Silicon | $-4.2 \times 10^{-6}$ | Tungsten | $6.8 \times 10^{-5}$ |

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## MAGNETIZATION CURRENT:




As discussed earlier, microscopic currents in materials can be modeled as equivalent surface currents for purpose of calculating magnetic dipole moments and magnetic fields.

Consider a magnetized cylinder of length $l$ and cross section area $A$ and magnetization $M$. Let $I_m$ be its surface current, then its magnetization can be related to its surface equivalent current as:

$$I_m A = M(Al) \Rightarrow \boxed{I_m = Ml}$$

→ Thus magnetisation can also be interpreted as equivalent surface current per unit length.

→ If a solenoid of N turns is completely filled with some material then if we want same magnetic field in absence of the magnetic material we will have to flow an additional current of $I_m/N$ (NCERT calls this additional current as magnetizing current).

NCERT EXAMPLE: 5.10:

**Example 5.10** A solenoid has a core of a material with relative permeability 400. The windings of the solenoid are insulated from the core and carry a current of 2A. If the number of turns is 1000 per metre, calculate (a) $H$, (b) $M$, (c) $B$ and (d) the magnetising current $I_m$.

**Solution**

(a) The field $H$ is dependent of the material of the core, and is
$H = nI = 1000 \times 2.0 = 2 \times 10^3$ A/m.

(b) The magnetic field $B$ is given by
$B = \mu_r \mu_0 H$
$= 400 \times 4\pi \times 10^{-7}\ (\text{N/A}^2) \times 2 \times 10^3$ (A/m)
$= 1.0$ T

(c) Magnetisation is given by
$M = (B - \mu_0 H)/\mu_0$
$= (\mu_r \mu_0 H - \mu_0 H)/\mu_0 = (\mu_r - 1)H = 399 \times H$
$\cong 8 \times 10^5$ A/m

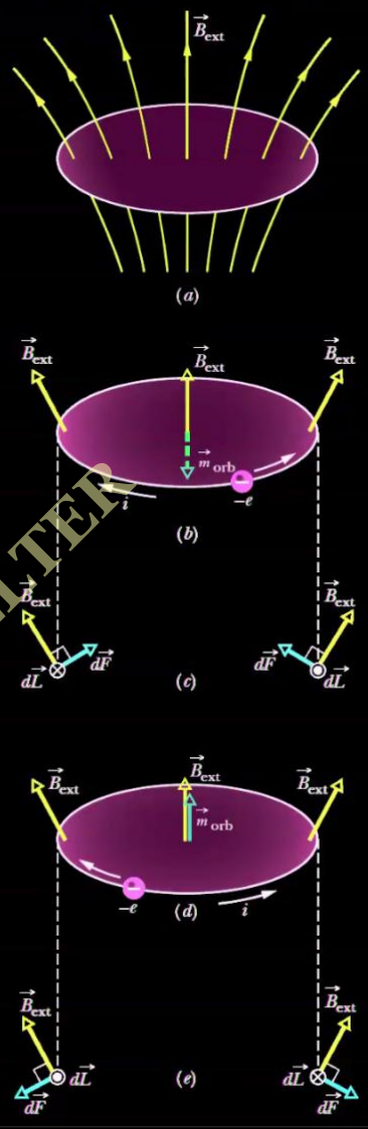
Alternatively:
$M = \chi H = 399 H$.

(d) The magnetising current $I_M$ is the additional current that needs to be passed through the windings of the solenoid in the absence of the core which would give a $B$ value as in the presence of the core. Thus $B = \mu_0 n (I + I_M)$. Using $I = 2$A, $B = 1$ T, we get $I_M = 794$ A.

(Alternatively, Magnetizing current $I_M = \frac{I_m}{N} = \frac{Ml}{N} = \frac{M}{n} = \frac{\chi H}{n}$)

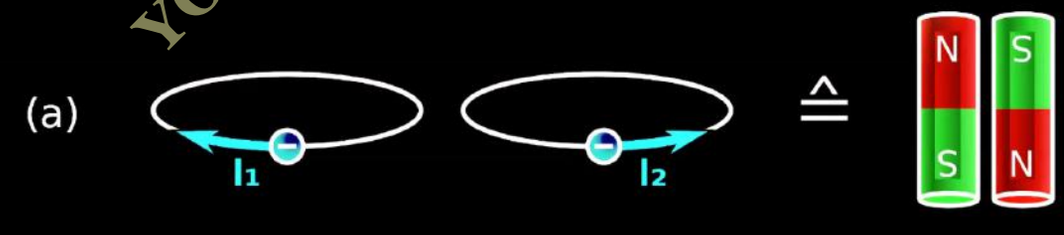
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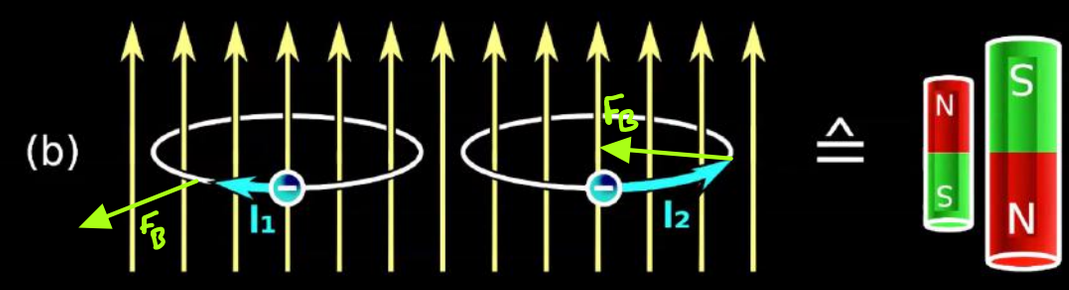
# Loop Model Of Electron Orbital Magnetic Moment In Non-Uniform Field:




While complete and correct explanation of magnetic behavior of materials requires understanding of quantum mechanics, we may also understand some magnetic behaviours of materials classically also. Here we can show how because of orbital magnetic moment, an atom may be attracted or repelled in a non uniform magnetic field. For simplicity we may assume a revolving electron to behave like a current loop. Now consider figure b i.e the case when orbital magnetic moment is opposite to the external field near pole of a bar magnet. Here considering an element of the current loop (two diametrically opposite elements) it can be seen that the loop experiences a net repulsion. Similarly, considering figure d i.e when magnetic moment is along the applied non-uniform field, there is a net resultant attractive force towards pole of the magnet. We will use this model in later sections to understand behaviors of paramagnetic and diamagnetic materials.

## Diamagnetism:

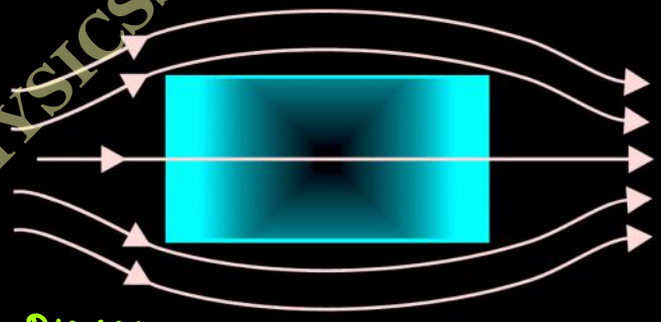



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Diamagnetic behaviour is exhibited by all atoms having paired electrons. Such materials produce a net magnetic moment opposite. Such atoms produce a net magnetic moment opposite to the applied field. To understand this consider the figure above. Due to opposite spins, the magnetic force on one of the electrons acts towards the centre thus increasing the net centripetal force (hence speeding it up). While on the other electron, the magnetic force acts outwards, thus slowing it down. The net result of this differential between the velocities of two electrons is that it creates a net magnetic moment opposite to that of applied field. Thus such materials when placed near poles of a bar magnet tend to be repelled (as described in the previous section on loop model of electron's orbital motion).

Since diamagnetic materials produce a magnetic moment opposite to the external field, the field inside such material is weaker than surroundings. The field lines for such a material are shown in adjacent figure.



DIAMAGNETIC SLAB IN UNIFORM EXTERNAL FIELD

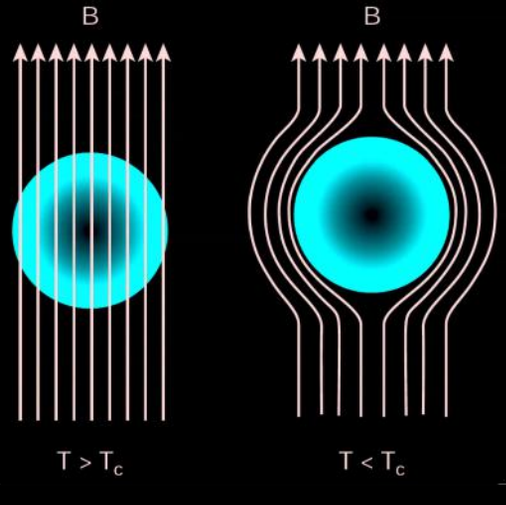
→ Diamagnetism is present in almost all materials, however in some of the materials it gets overshadowed by other effects such as paramagnetism and ferromagnetism etc.

→ Apart from materials listed in the table in a previous section, water and sodium chloride are also diamagnetic.

→ In such materials susceptibility is almost temperature independent.

## MEISSNER EFFECT :

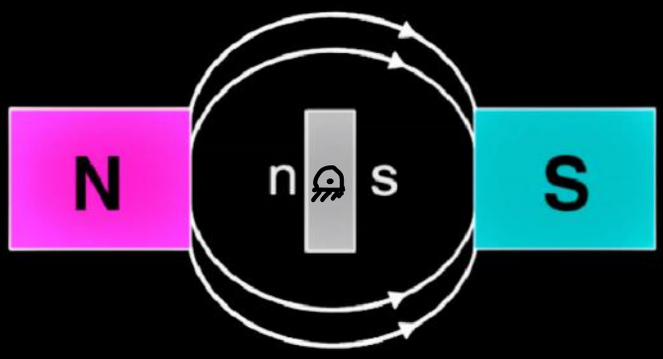
Superconductors are a special variety of diamagnetic materials whose susceptibility $\chi = -1$. Such materials do not allow any magnetic field line to pass through them and this phenomenon is called Meissner effect. Typically metals behave like superconductors when cooled below a certain critical temperature $T_c$.



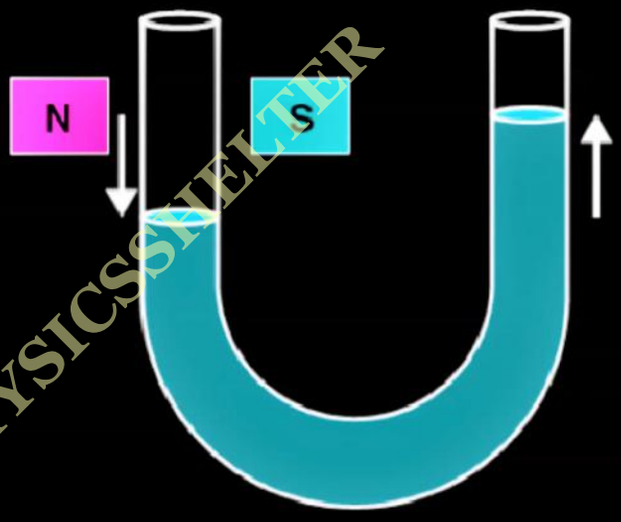

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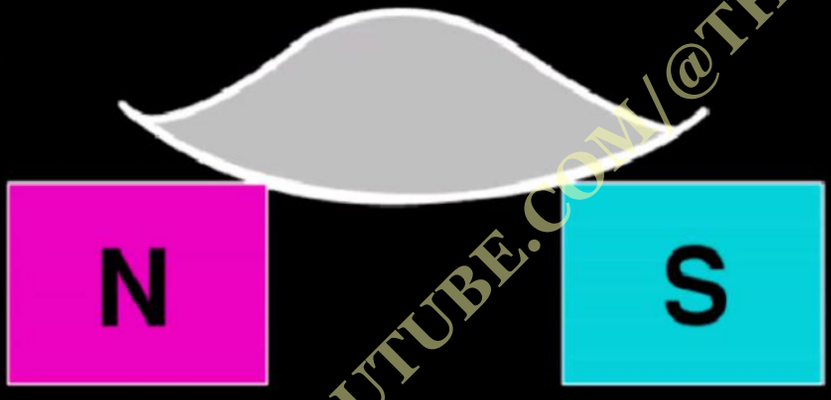
## Some More Properties Of Diamagnetic Materials:

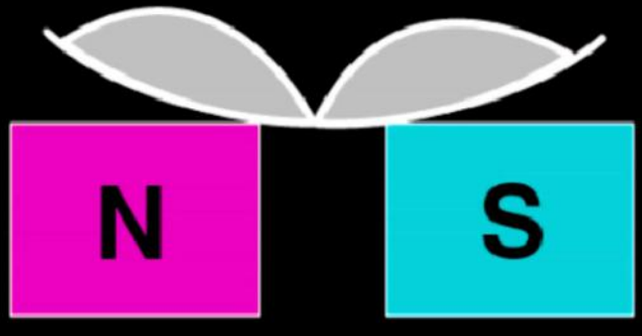
→ Diamagnetic materials tend to accumulate in regions of weaker magnetic fields thus if a pivoted diamagnetic rod is free to rotate between poles of a U magnet, it orients itself perpendicular to the poles.




→ If a diamagnetic liquid is filled in a U tube and one of the limbs is sandwitched between poles of a magnet, the liquid rises in the other limb and falls in the limb between poles of a magnet (see figure).




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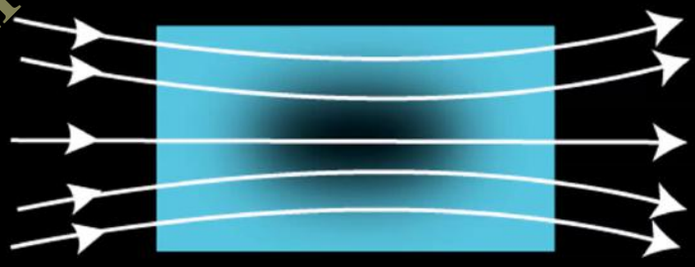
→ If a diamagnetic liquid is taken in a watch glass and placed between poles of a magnet then if poles are sufficiently apart, liquid develops a bulge in the middle (region of weak field) while if the poles are sufficiently close, it develops a depression in between. (Since field is strong there).

→ If a frog is placed in sufficiently strong field created by currents in a vertical solenoid, it may levitate. This is because of water present in frog's body which is diamagnetic. The repulsion due to non-uniform field created by the solenoid can balance the weight of the frog.



An overhead view of a frog that is being levitated in a magnetic field produced by current in a vertical solenoid below the frog.

**PARAMAGNETISM:** In certain atoms net resultant of spin and orbital magnetic moments is not zero, thus the atoms get permanent magnetic dipole moment. When such an atom is subjected to an external field, it experiences a magnetic torque and achieves a stable configuration when its magnetic moment becomes parallel to the applied field thus in such materials, net magnetic field is more than the applied field because the field due to magnetisation of the material is in the same direction as the applied field. The adjacent figure shows the field line pattern in a paramagnetic bar put in an external uniform field.



Magnetic Field lines through Paramagnetic material

**CURIE'S LAW:** When a paramagnetic salt is subjected to an external field at certain temperature, two competing phenomena take place simultaneously. The external magnetic field tends to align the magnetic dipoles while thermal agitation tends to randomize their directions. Thus with increase in temperature, the material achieves a smaller magnetization for the same value of applied field. Curie's law describes this phenomenon quantitatively and is given by:

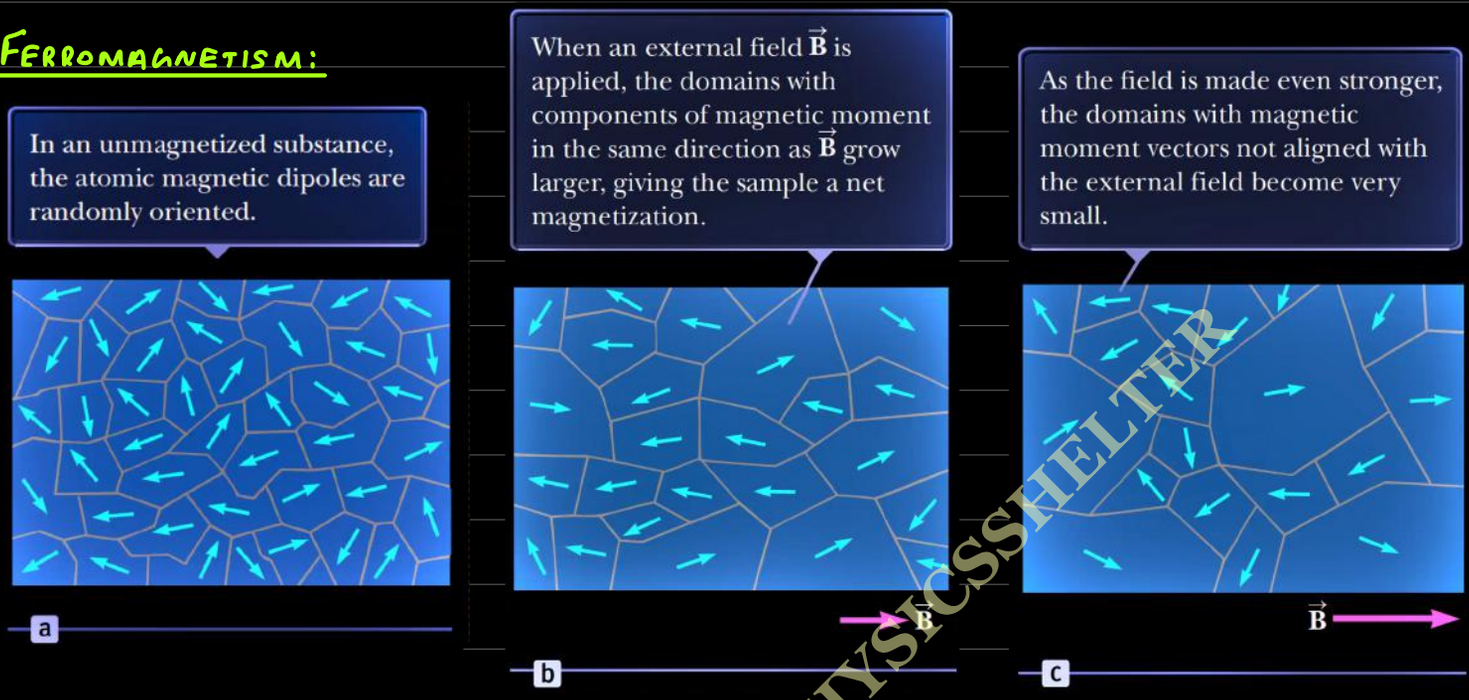
$$M = C B_0 / T$$

$B_0$: Applied Field  T: Temperature  C: Curie's Constant
M: Magnetization (Curies Constant depends on material).

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Curie's law is only an approximate law and is valid only when $B_0/T$ is not very large. If $B_0/T$ is very large, the exact analysis can be done using Quantum theory.

## FERROMAGNETISM:



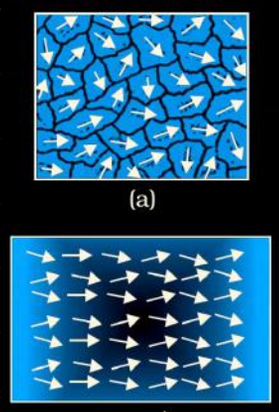

In all ferromagnetic substances there are microscopic regions known as domains within which all magnetic moments are perfectly aligned even though domains themselves may have random orientations. These domains have volumes about $10^{-12}$ to $10^{-8}$ $m^3$ and contain about $10^{17}$ to $10^{21}$ atoms. The domain structure occurs due to a quantum mechanical phenomenon known as exchange coupling.

The boundaries between domains of different orientations are called domain walls. In an unmagnetized sample the magnetic domains are randomly oriented so that net magnetic moment is zero. When the sample is placed in external magnetic field, the domains with magnetic moments aligned (or acute angle) with field grow in size while those not aligned grow smaller. If the external field becomes very strong, the unfavorably oriented domains will become extremely small.

Besides the phenomenon of domain growth and decline, there may also occur the phenomenon of domain realignment in which a domain may change its alignment completely.

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When the external field is removed, sample may retain a net magnetization in the direction of original field. This is because the domains once aligned will not completely become randomized upon removal of external field, although randomisation may be achieved either by heating the magnetized material to a sufficiently high temperature or through mechanical impact (say hammering or dropping the magnetized block).



(a) Randomly oriented domains, (b) Aligned domains.

**CURIE TEMPERATURE:** If the temperature of ferromagnetic substance is increased, there comes a point where the domain structure ceases to exist and the thermal agitations are sufficient to randomize individual atomic magnetic dipoles. The temperature at which this happens is called the Curie Temperature. Beyond Curie Temperature, the ferromagnetic substances begin to behave just like Paramagnetic Substances.

**CURIE'S LAW (CURIE WEISS LAW):**

At temperatures sufficiently greater than Curie's Temperature, the susceptibility of any ferromagnetic substance is given as:

$$\chi = C/(T - T_c)$$

$T > T_c$. This expression is not valid near $T = T_c$ and at very very high temperatures, $T_c$ is replaced by another temperature. The Curie's law is only an approximate law.

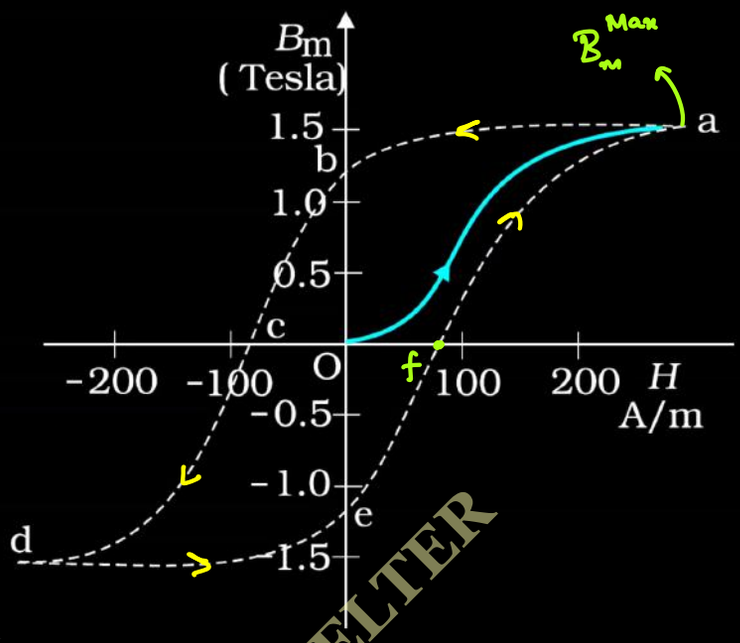
**CURIE TEMPERATURE $T_c$ OF SOME FERROMAGNETIC MATERIALS**

| Material | $T_c$ (K) |
|---|---|
| Cobalt | 1394 |
| Iron | 1043 |
| $Fe_2O_3$ | 893 |
| Nickel | 631 |
| Gadolinium | 317 |

→ Due to domain structures, susceptibility of ferromagnetic materials is very high (typically few hundreds to few thousands).

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## MAGNETIC HYSTERESIS CURVE:




The adjacent figure shows a variation of $B_m$ (due to magnetization) and the magnetizing field (H) for a ferromagnetic material. The relation between $B_m$ and H for ferromagnetic materials is complex. It is often non-linear and depends on the magnetic history of the sample. In the shown figure one can trace one cycle of magnetization and demagnetization of a ferromagnetic sample. There a certain technical terms that we need to know about the curve.

## SATURATION MAGNETIZATION:

As the magnetizing field (H) is increased, there comes a point when the material is almost completely magnetized. The value of magnetisation at this point is called saturation magnetization. For our curve, the saturation magnetization is given as $M_{max} = B_m^{Max} / \mu_0 = 1.5 \text{ Tesla}/\mu_0$. (This corresponds to point a in the curve).

## RETENTIVITY (REMANENCE):

As the magnetizing field is reduced from point of saturation (a), the magnetization also keeps reducing (though follows a different curve from the initial curve for magnetizing). This is shown along the segment a-b of the curve. It is seen that even when the magnetizing field is removed, still there remains some magnetization in the material. The value of $B_m$ corresponding to this state is called retentivity (point b in the curve).

**COERCIVITY:** In order to make the net magnetisation of a sample zero (bringing back from saturation magnetization), a certain amount of reverse magnetizing field is required (point c in the curve). The value of H at this point is called coercivity.

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→ Once the reversed H is increased more and more beyond point of coercivity, the will come a point of reverse saturation (d). If now reversed H is gradually decreased once again we will reach the point of retentivity (e) when H becomes zero. When H again increases in forward direction once again we will reach the point of coercivity (f) and continuing further we will again reach a the completing the cycle.

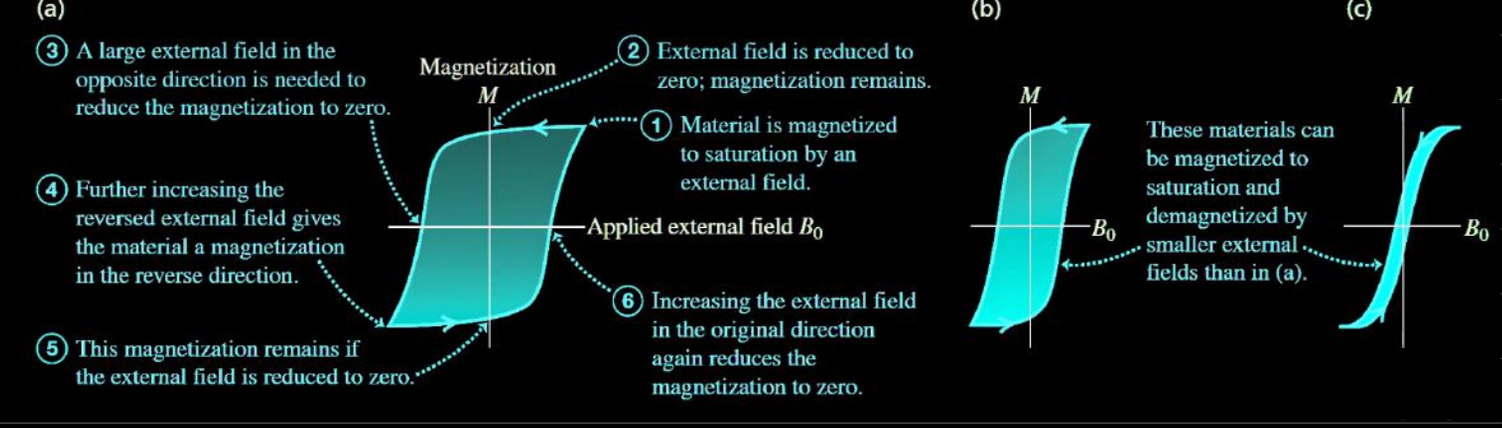
→ Area enclosed in the hysteresis curve is proportional to the total heat loss in the process of magnetizing and demagnetizing a material. This becomes especially important while dealing with materials undergoing repeated cycles of magnetizing and demagnetizing such as Iron cores in transformers or choke coils in Alternating current conditions.

## Soft Vs Hard Ferromagnets:

If upon removal of external field significant amount of magnetization persists in a material, it is called a hard ferromagnet. Alnico (Alloy of Iron, Nickel, Cobalt and Aluminum) is one such alloy. Another example of such substance is naturally occuring lodestone. Such materials are suitable for making permanent magnets (and compass needles). On the other hand if upon removal of external field, the magnetization disappears, we call such materials as soft ferromagnets (soft iron is one such material).

## Permanent Magnets And Electromagnets:

Hysteresis loops: The materials of both (a) and (b) remain strongly magnetized when $B_0$ is reduced to zero. Since (a) is also hard to demagnetize, it would be good for permanent magnets. Since (b) magnetizes and demagne-tizes more easily, it could be used as a computer memory material. The material of (c) would be useful for transformers and other alternating-current devices where zero hysteresis would be optimal.




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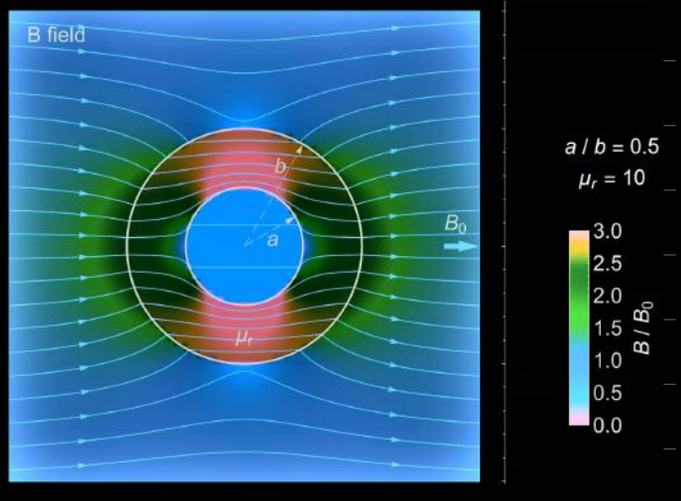
Hysterusis curves enable us to select suitable meteriels for various purpose magnets (see figure above).

## PERMANENT MAGNETS:

The meterial for permanent magnets should have high retentivity (to produce strong magnetic fields) and high coercivity (so that magnetic field is not erased by temperature fluctuations, stray magnetic fields or minor mechanical impacts). In addition, the meterial for permanent magnets should have high permeability. The meterials used for making permanent magnets includes Steel, Alnico, Cobalt and Ticonal.

**ELECTROMAGNETS:** The meterials used for electromagnets should have high permeability and low retentivity (also a slim hysterusis curve because area in a hysterusis curve is proportional to the heat lost in magnetizing demagnetizing cycles). Typical meterial used for making electromagnets is soft iron. Electromagnets are used as transformer cores, solenoid cores and used in making electric bells, loudspeakers, telephone diaphragms and cranes to lift machinery and bulk quantities of iron and steel.

**MAGNETIC SHIELDING:** To protect sensitive equipment from magnetic fields, we can keep them in thick cavities made of soft iron. The soft iron lets pass most of the magnetic field lines through its metrix, thus protecting anything in its cavity from external strong magnetic fields.




**NCERT EXAMPLE: 5.11:**

**Example 5.11** A domain in ferromagnetic iron is in the form of a cube of side length 1μm. Estimate the number of iron atoms in the domain and the maximum possible dipole moment and magnetisation of the domain. The atomic mass of iron is 55 g/mole and its density is 7.9 g/cm$^3$. Assume that each iron atom has a dipole moment of $9.27\times10^{-24}$ A m$^2$.

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**Solution** The volume of the cubic domain is

$V = (10^{-6}\ \text{m})^3 = 10^{-18}\ \text{m}^3 = 10^{-12}\ \text{cm}^3$

Its mass is volume × density $= 7.9\ \text{g cm}^{-3} \times 10^{-12}\ \text{cm}^3 = 7.9 \times 10^{-12}\ \text{g}$

It is given that Avagadro number ($6.023 \times 10^{23}$) of iron atoms have a mass of 55 g. Hence, the number of atoms in the domain is

$$N = \frac{7.9 \times 10^{-12} \times 6.023 \times 10^{23}}{55}$$

$$= 8.65 \times 10^{10}\ \text{atoms}$$

The maximum possible dipole moment $m_{max}$ is achieved for the (unrealistic) case when all the atomic moments are perfectly aligned. Thus,

$m_{max} = (8.65 \times 10^{10}) \times (9.27 \times 10^{-24})$

$= 8.0 \times 10^{-13}\ \text{A m}^2$

The consequent magnetisation is

$M_{max} = m_{max}/\textit{Domain volume}$

$= 8.0 \times 10^{-13}\ \text{Am}^2/10^{-18}\ \text{m}^3$

$= 8.0 \times 10^{5}\ \text{Am}^{-1}$

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